SEA 820 NLP FINAL PROJECT

Detecting AI-generated text

HUMAN
OR
AI?

TF-IDF Logistic Regression vs. fine-tuned DistilBERT

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Detection is useful only if its uncertainty is understood

Research question

Does fine-tuned DistilBERT outperform a transparent TF-IDF baseline on the same held-out test set?

Why it matters

Academic integrity

Content moderation

Misinformation response

Risk of false accusation

A shared test set makes the comparison meaningful

1

Kaggle dataset
AI vs. Human Text

2

Clean + stratify
70 / 15 / 15

3

Train two models
Classic + Transformer

4

Evaluate
Metrics + errors

Default final run: 12,000 stratified samples | seed 42 | AI is the positive class

The models test feature engineering against transfer learning

Classic baseline

TF-IDF

Unigrams + bigrams
100,000-feature cap
Class-balanced Logistic Regression

Transformer

DistilBERT

256-token maximum
Batch size 16 | learning rate 2e-5
Best validation-F1 checkpoint

Results will identify whether added complexity pays off

MODEL	ACCURACY	PRECISION	RECALL	F1
TF-IDF + Logistic Regression	0.9828	0.9908	0.9627	**0.9765**
DistilBERT	0.9778	0.9578	0.9836	**0.9705**

The TF-IDF baseline achieved the highest F1-score, outperforming DistilBERT by 0.60 percentage points. DistilBERT achieved higher recall but produced more false positives.

The most important errors are confident false accusations

Error analysis

29 false positives
11 false negatives
High recall came at the cost of incorrectly flagging more human writing.

Responsible-use limit

Never treat a prediction as proof
Require human review and appeal
Evaluate bias across writer groups
Expect errors on unseen AI models

A detector is evidence to review, not a verdict

TF-IDF Logistic Regression was the best overall model: 98.28% accuracy and 97.65% F1.

What we learned

1. A fair comparison needs one held-out test set.

2. F1 and error types matter more than accuracy alone.

3. Generalization and fairness remain unresolved.